



Problems (Relations and Functions)

1. If $A = \{-1, 1\}$, find $A \times A \times A$
2. Let $A = \{1, 2\}$, $B = \{1, 2, 3, 4\}$, $C = \{5, 6\}$ and $D = \{5, 6, 7, 8\}$. Verify that (i) $A \times (B \cap C) = (A \times B) \cap (A \times C)$. (ii) $A \times C$ is a subset of $B \times D$
3. Write out the indicated sets by listing their elements between braces. Suppose $A = \{1, 2, 3, 4\}$ and $B = \{a, c\}$
 - a. $A \times B$
 - b. $A \times A$
 - c. $\phi \times B$
4. Sketch the following
 - a. $[0, 1] \times [0, 1]$
 - b. $\mathbb{N} \times \mathbb{Z}$
 - c. $[0, 1] \times \{1\}$
 - d. $[0, 1] \times [0, 1] \times [0, 1]$
 - e. $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\} \times [0, 1]$
 - f. $\{x \in \mathbb{R} : x^2 = 2\} \times \{x \in \mathbb{R} : |x| = 2\}$
5. Consider the relation $R = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a)\}$ on set $A = \{a, b, c, d\}$. Is R reflexive? Symmetric? Transitive? If a property does not hold, say why.
6. There are two different equivalence relations on the set $A = \{a, b\}$. Find them
7. A function $f: \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ is defined as $f(n) = (2n, n+3)$. Verify whether this function is injective and whether it is surjective.
8. Prove that the function $f: \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{5\}$ defined by $f(x) = (5x+1)/(x-2)$ is bijective
9. Define a relation R on the set $\mathbb{N}$ of natural numbers by $R = \{(x, y): y = x + 5, x \text{ is a natural number less than } 4; x, y \in \mathbb{N}\}$. Depict this relationship using roster form. Write down the domain and the range.
10. Determine the domain and range of the relation R defined by $R = \{(x, x + 5): x \in \{0, 1, 2, 3, 4, 5\}\}$
11. Write the relation $R = \{(x, x^3) : x \text{ is a prime number less than } 10\}$ in roster form
12. Let R be the relation on $\mathbb{Z}$ defined by $R = \{(a, b): a, b \in \mathbb{Z}, a - b \text{ is an integer}\}$. Find the domain and range of R
13. Find the domain and range of the following real functions: (i) $f(x) = -|x|$ (ii) $f(x) = \sqrt{9 - x^2}$
14. The relation f is defined by $f(x) = \begin{cases} x^2, & 0 \leq x \leq 3 \\ 3x, & 3 \leq x \leq 10 \end{cases}$ and the relation g is defined by $g(x) = \begin{cases} x^2, & 0 \leq x \leq 2 \\ 3x, & 2 \leq x \leq 10 \end{cases}$. Show that f is a function and g is not a function.

15. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be defined, respectively by $f(x) = x + 1$, $g(x) = 2x - 3$. Find $f + g$, $f - g$ and f/g
16. Consider the functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = 1 \times 2 + 1$ and $g(x) = 3x + 2$. Find the formulas for $g \circ f$ and $f \circ g$
17. Let $f = \{(1,1), (2,3), (0, -1), (-1, -3)\}$ be a function from $\mathbb{Z}$ to $\mathbb{Z}$ defined by $f(x) = ax + b$, for some integers a, b . Determine a, b .
18. Let f be the subset of $\mathbb{Z} \times \mathbb{Z}$ defined by $f = \{(ab, a + b): a, b \in \mathbb{Z}\}$. Is f a function from $\mathbb{Z}$ to $\mathbb{Z}$? Justify your answer
19. Let $B = \{2^n : n \in \mathbb{Z}\}$. Show that the function $f: \mathbb{Z} \rightarrow B$ defined as $f(n) = 2^n$ is bijective. Then find f^{-1}
20. Consider the function $f: \{1,2,3,4,5,6,7\} \rightarrow \{0,1,2,3,4,5,6,7,8,9\}$ given as $f = \{(1,3), (2,8), (3,3), (4,1), (5,2), (6,4), (7,6)\}$. Find: $f(\{1,2,3\})$, $f(\{4,5,6,7\})$, $f(\emptyset)$, $f^{-1}(\{0,5,9\})$ and $f^{-1}(\{0,3,5,9\})$